Name
Date
Course

Probability (Homework)

① An aviary has 300 parakeets. 180 have white feathers, 80 have blue feathers, and 50 have white feathers and blue feathers.

i) How many parakeets have white feathers but no blue feathers?

$180-30=\boxed{150}$

ii) How many parakeets have blue feathers but no white feathers?

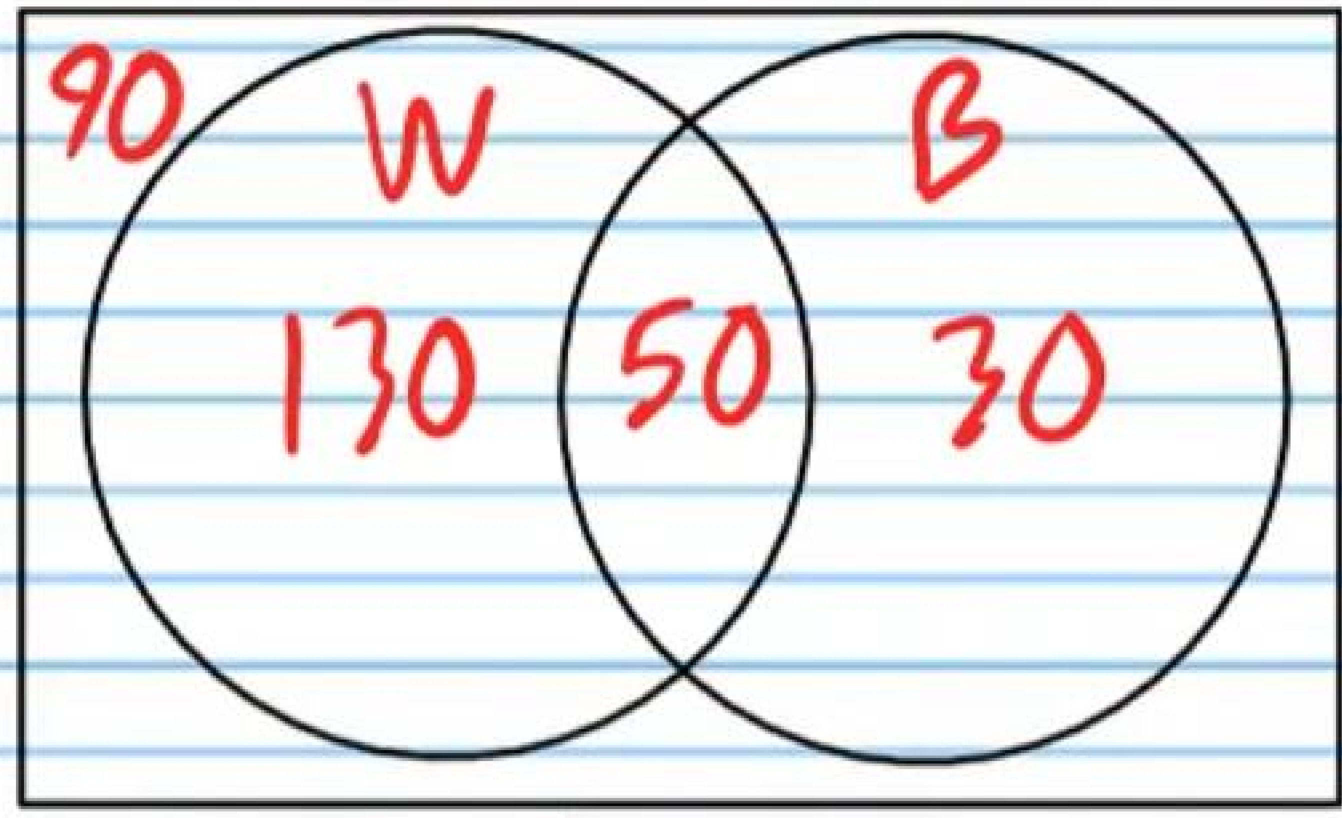
$80-50=\boxed{30}$

iii) How many parakeets have either white or blue feathers?

$130+50+30=\boxed{210}$

iv) How many parakeets have no white or blue feathers?

$300-210=\boxed{90}$



v) What is the probability that a randomly selected parakeet will have white feathers?

vi) What is the probability that a randomly selected parakeet will have blue feathers?

vii) What is the probability that a randomly selected parakeet will have neither white feathers nor blue feathers?

viii) What is the probability that a randomly selected parakeet will have both white feathers and blue feathers?

ix) What is the probability that a randomly selected parakeet will have either white feathers or blue feathers?

② A tech factory makes 100 calculators one day. 17 of the calculators are nonfunctional, 21 of them have scratches, and 10 of them are nonfunctional with scratches.

i) How many calculators are functional but have scratches?

$21-10=\boxed{11}$

ii) How many calculators are nonfunctional but have no scratches?

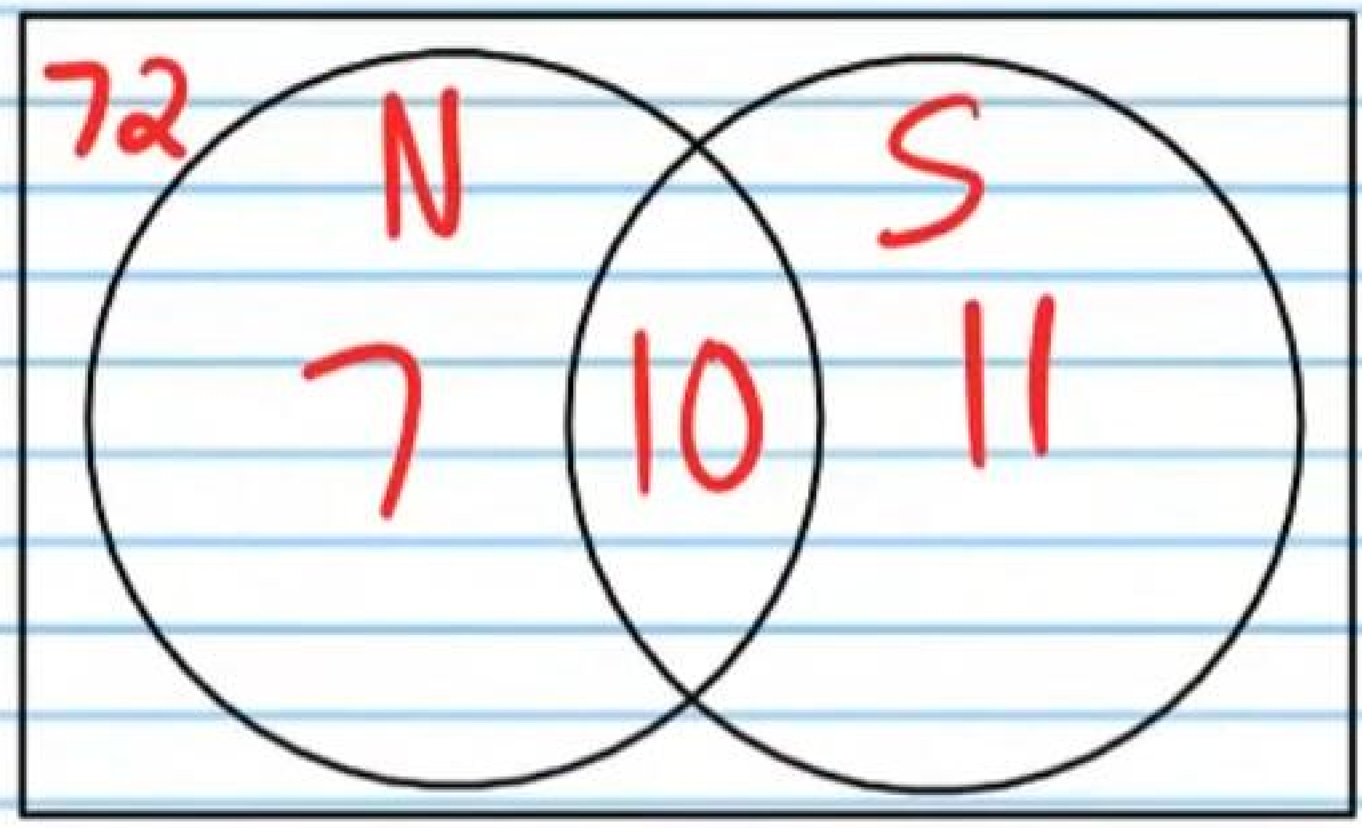
$17-10=\boxed{7}$

iii) How many calculators have either scratches or are nonfunctional?

$7+10+11=\boxed{28}$

iv) How many calculators have no stratches and are operational?

$100-28=\boxed{72}$



v) What is the probability that a randomly chosen calculator will be nonfunctional?

vi) What is the probability that a randomly chosen calculator will have scratches?

vii) What is the probability that a randomly chosen calculator will be neither nonfunctional nor have scratches?

viii) What is the probability that a randomly chosen calculator will be both nonfunctional and have scratches?

ix) What is the probability that a randomly chosen calculator will be either nonfunctional or have scratches?

(3) The probability that a randomly selected car will be a Toyota is $3/25$. The probability that a randomly selected car will be a Ford is $2/25$. What is the probability that a randomly selected car will be either a Toyota or a Ford?

(4) The probability that a randomly chosen restaurant will be Taco Bell is 16%. The probability that a randomly chosen restaurant will be Del Taco is 7%. What is the probability that a randomly chosen restaurant will be either Taco Bell or Del Taco?

(5) A soda drink is randomly selected. The probability that the soda will be a Coca-Cola is 0.21. The probability that the soda will be a Pepsi is 0.09. What is the probability that a randomly selected soda will be either a Coca-Cola or a Pepsi?

(6) The probability that a randomly selected car will be a Toyota is $3/25$. The probability that a randomly selected car will have ABS brakes is $9/10$. The probability that a randomly selected car will both be a Toyota and have ABS brakes is $1/10$. What is the probability that a randomly selected car will be either a Toyota or have ABS brakes?

(7) The probability that a family will eat at Taco Bell is 16%. The probability that the same family will order in the drive thru is 61%. The probability that this family will do both is 9%. What is the probability that this family will eat at either Taco Bell or order in the drive thru?

(8) A soda drink is randomly selected. The probability that the soda will be a Coca-Cola is 0.21. The probability that the soda will be cherry flavored is 0.05. The probability that the soda will be both a Coca-Cola and cherry flavored is 0.02. What is the probability that a randomly selected soda will be either a Coca-Cola or cherry flavored?

⑨ There are 5 blue marbles, 6 green marbles, and 8 red marbles in a bag. 3 marbles are chosen at random.

i) What is the probability that all 3 are blue?

ii) What is the probability that all 3 are red?

iii) What is the probability that all 3 are green?

iv) What is the probability that 2 marbles will be red and 1 will be blue?

v) What is the probability that 2 will be green and 1 will be blue?

vi) What is the probability that 2 will be blue and 1 will be red?

vii) What is the probability that 2 will be blue and 1 will be green?

viii) What is the probability that 1 marble will be green, 1 will be blue, and 2 will be red?

(10) Bobby has a coin collection. He has 5 pennies, 6 dimes, and 9 quarters. He wants to pick 4 coins at random to show his fellow classmates.

i) What is the probability that all 4 are pennies?

ii) What is the probability that all 4 are dimes?

iii) What is the probability that all 4 are quarter?

iv) What is the probability that 3 coins will be pennies and 1 will be a dime?

v) What is the probability that 3 will be dimes and 1 will be a quarter?

vi) What is the probability that 2 will be quarters and 2 will be pennies?

vii) What is the probability that 2 will be dimes and 2 will be pennies?

viii) What is the probability that 1 coin will be a penny, 1 will be a dime, and 2 will be quarters?

(11) A boy attempts to throw a ball into a basket. The probability that the boy will miss the basket is 38%. What is the probability that the ball will land in the basket?

(12) The probability that Laurie will lose her phone is 11%. What is the probability that she will not lose her phone?

⑬ Amy is going to arrange flowers in a small bouquet. She will choose 4 flowers at random. There are 8 pink flowers and 6 white flowers to choose from.

i) What is the probability that she will choose at least one pink flower?

ii) What is the probability that she will choose at least one white flower?

Homework Answers

1. v) $\boxed{\frac{3}{5}}$ or $\boxed{0.6}$ or $\boxed{60\%}$

vi) $\boxed{\frac{4}{15}}$ or $\boxed{0.2\dot{6}}$ or $\boxed{26.\dot{6}\%}$

vii) $\boxed{\frac{3}{10}}$ or $\boxed{0.3}$ or $\boxed{30\%}$

viii) $\boxed{\frac{1}{6}}$ or $\boxed{0.1\dot{6}}$ or $\boxed{16.\dot{6}\%}$

ix) $\boxed{\frac{7}{10}}$ or $\boxed{0.7}$ or $\boxed{70\%}$

2. v) $\boxed{\frac{17}{100}}$ or $\boxed{0.17}$ or $\boxed{17\%}$

vi) $\boxed{\frac{21}{100}}$ or $\boxed{0.21}$ or $\boxed{21\%}$

vii) $\frac{72}{100} = \boxed{\frac{18}{25}}$ or $\boxed{0.72}$ or $\boxed{72\%}$

viii) $\frac{10}{100} = \boxed{\frac{1}{10}}$ or $\boxed{0.1}$ or $\boxed{10\%}$

ix) $\frac{28}{100} = \boxed{\frac{7}{25}}$ or $\boxed{0.28}$ or $\boxed{28\%}$

3. $\boxed{0.2}$ or $\boxed{20\%}$ or $\boxed{1/5}$

4. $\boxed{0.23}$ or $\boxed{23\%}$ or $\boxed{23/100}$

5. $\boxed{0.3}$ or $\boxed{30\%}$ or $\boxed{\frac{3}{10}}$

6. $\boxed{0.92}$ or $\boxed{92\%}$ or $\boxed{\frac{23}{25}}$

7. $\boxed{0.68}$ or $\boxed{68\%}$ or $\boxed{\frac{17}{25}}$

8. $\boxed{0.24}$ or $\boxed{24\%}$ or $\boxed{\frac{6}{25}}$

9. i) $\boxed{\frac{10}{969}}$

ii) $\boxed{\frac{56}{969}}$

iii) $\boxed{\frac{20}{969}}$

iv) $\boxed{\frac{140}{969}}$

v) $\boxed{\frac{25}{323}}$

vi) $\boxed{\frac{80}{969}}$

vii) $\boxed{\frac{20}{323}}$

viii) $\frac{840}{969} = \boxed{\frac{280}{323}}$

10 i) $\frac{5}{4845} = \boxed{\frac{1}{969}}$

ii) $\frac{15}{4845} = \boxed{\frac{1}{323}}$

iii) $\frac{126}{4845} = \boxed{\frac{42}{1615}}$

iv) $\frac{60}{4845} = \boxed{\frac{4}{323}}$

v) $\frac{180}{4845} = \boxed{\frac{12}{323}}$

vi) $\frac{360}{4845} = \boxed{\frac{24}{323}}$

vii) $\frac{150}{4845} = \boxed{\frac{10}{323}}$

viii) $\frac{1080}{4845} = \boxed{\frac{72}{323}}$

11 $\boxed{62\%}$ or $\boxed{0.62}$ or $\boxed{\frac{31}{50}}$

12 $\boxed{89\%}$ or $\boxed{0.89}$ or $\boxed{\frac{89}{100}}$

13 i) $\boxed{\frac{15}{1001}}$

ii) $\frac{70}{1001} = \boxed{\frac{10}{143}}$